WITSANUPONG KOLAKUL

AI Engineer & Researcher · Computer Vision · NLP · Space Technology

Saraburi & Bangkok, Thailand teverygood@gmail.com github.com/itertius linkedin.com/in/witsanupong-kolakul

EDUCATION

Chulalongkorn University — Bangkok, Thailand *2025 – Present*
B.Eng. Computer Engineering and Digital Technology (CEDT)
Year 1, Class of 2030 · Focus: Artificial Intelligence, Computer Vision, DevOps, Space Technology

Saraburi Wittayakhom School — Saraburi, Thailand *2020 – March 2026*
High School Diploma · Science–Mathematics Program · GPAX 3.59
Advanced Mathematics, Physics, Chemistry, and Biology

RESEARCH & PUBLICATIONS

Perceptual Loss-Based Approaches to Personal Color Season Classification
CVPR 2026 · Non-archival Track (4 pages)
Two-stage pipeline with hybrid cross-entropy + CIEDE2000 perceptual loss for seasonal color classification. Reduces cross-undertone errors by 4.2% (±0.42%) and improves accuracy by 13.4% compared to baseline (46.9% ±0.42%).

AWARDS & COMPETITIONS

Best Deployment **Thailand CANSAT–ROCKET Competition 2025** *2025*
NSM · Defence Technology Institute · 104 teams · Prize: 5,000 THB
PM & Software Engineer. Ground Station, Flight Software, and Deployment Software. Only team in Thailand with this end-to-end system design.

1st Runner-Up **OIC InsureTech Award 2025** *2025*
Center of InsureTech Thailand · 400+ teams · Prize: 70,000 THB
"InsureMate" — AI application for insurance literacy integrating ML and NLP/LLM features.

Silver Medal **Thailand AI Engineering Championship 2025** *2025*
CMKL University · AST · 800+ participants
Topics: KNN, K-Means Clustering, DAG Scheduling. 2nd place after an intensive 2-month preparation phase.

1st Runner-Up **DeafTech Hackathon 2024** *2024*
UNICEF Thailand · Prize: 6,000 THB
"ThaiSign Application" — Thai sign language learning app using MediaPipe + MobileNetV2.

Outstanding AI **AI Builders 5 — Tomato Disease Classification** *2024*
Mahidol University · Central Retail Digital
Edge contour detection for plant separation; addressed data imbalance and real-world noise.

Consolation **Thailand CANSAT–ROCKET Competition 2024** *2024*
NSM · Defence Technology Institute · 100 teams
Software Engineer & Systems Integration. Advanced to national finals.

Semi-Finalist **Thailand Metaverse Hackathon #1 & #2** *2024 – 2025*
Faculty of Engineering, Chulalongkorn University · OIC Thailand · Top 20 teams
PM & Full-Stack Developer. VR app for OCD Exposure Therapy (Unity, React Native, Node.js); VR insurance learning simulation.

1st Runner-Up **Kasetsart Water Rocket Championship 2024** *2024*
Water rocket design and flight optimization competition. Achieved 1st Runner Up through innovative aerodynamic design and precise trajectory calculations.

SPU AI PROMPT Mini Hackathon 2025 *2025*
EDA, data cleaning, and AutoGluon model training for prompt quest challenge. Top quartile performance.

SELECTED PROJECTS

AI for Thai — Exercise Form Analysis
MediaPipe · Pathumma LLM · VAJA TTS
Real-time AI coaching with pose detection and Thai TTS feedback for athletes.

AI Thai Learning Companions
NLP · LLM · Text-to-Speech
AI-powered Thai learning app with vocabulary exercises and TTS functionality.

Subscription Hub — Payment Tracker
React · Node.js · Docker · GitHub Actions
Full-stack payment tracker with multi-currency support and CI/CD deployment.

BostonHousing - Smart Regression
Regression · Data Visualization · Analytics
Data cleaning, feature engineering, and evaluation with scikit-learn and pandas.

TECHNICAL SKILLS

Machine Learning: *PyTorch, TensorFlow, Scikit-learn, AutoGluon, Optuna, XGBoost*

Languages: *Python, C, C++, JavaScript*

Computer Vision: *FaRL, CLIP, OpenCV, MediaPipe, EasyOCR, Tesseract*

Frameworks: *React, Vue.js, Node.js, NumPy, Pandas, Matplotlib, Plotly*

NLP & LLMs: *Thai NLP, BM25, TF-IDF, RAG Pipelines, Typhoon, OpenThaiGPT*

Tools & Platforms: *Git, Docker, Google Colab, Kaggle, Figma, Unity, LaTeX, Arduino*

VOLUNTEER & LEADERSHIP

Team Mentor & Advisor — Thailand CANSAT–ROCKET 2026 *2026*
Volunteer PM for two finalist teams (Sky's Line & Sky's Rift 2). Technical guidance throughout competition cycle.

Guest Speaker — OIC International Insurance Symposium 2025 *2025*
Invited speaker on AI applications in insurance technology and innovation.

Teaching Assistant — Computer Olympiad Training Camp *2025*
Algorithms, data structures, and competitive programming in C++.

President of Yellow House — Saraburi Wittayakhom School *2024*
Managed 1 of 4 houses with 1000+ members. Led house to victory as best house during SBW games week.

Technical Writing (Medium / GitHub) *Ongoing*
Published technical articles and tutorials on AI, machine learning, and software development.

References available upon request · Last updated April 2026